CBSE Practice Worksheet

Class 8 | Mathematics | Difficulty: Hard | Set A

Instructions: Answer all questions. This worksheet contains 25 questions including multiple choice, fill in the blanks, and short answer questions.

Section A: Multiple Choice Questions (1 mark each)

1. Find HCF of $(x^2 - 4)$ and $(x^2 - 4x + 4)$
   - (A) $x - 2$
   - (B) $x + 2$
   - (C) $(x-2)^2$
   - (D) $x^2 - 4$
2. Factorize: $x^2 - 5x - 14$
   - (A) (x-7)(x+2)
   - (B) (x+7)(x-2)
   - (C) (x-7)(x-2)
   - (D) (x+7)(x+2)
3. Solve: 2^(x+1) = 32
   - (A) x = 3
   - (B) x = 4
   - (C) x = 5
   - (D) x = 6
4. If $x + 1/x = 5$, find $x^2 + 1/x^2$
   - (A) 21
   - (B) 23
   - (C) 25
   - (D) 27
5. Simplify: $(a^2 - b^2)/(a + b)$
   - (A) a - b
   - (B) a + b
   - (C) $a^2 - b$
   - (D) $a - b^2$
6. If a:b = 3:4 and b:c = 5:6, find a:c
   - (A) 3:6
   - (B) 5:8
   - (C) 15:24
   - (D) 1:2
7. If 2■ = 3■ = 6■, which is true?
   - (A) 1/z = 1/x + 1/y
   - (B) z = x + y
   - (C) z = xy
   - (D) z = x - y
8. Simplify: $(x^3 - 8)/(x - 2)$
   - (A) $x^2 + 2x + 4$
   - (B) $x^2 - 2x + 4$
   - (C) $x^2 + 4$

(D) $x^2 - 4$

9. What is the cube root of -125?
   (A) 5
   (B) -5
   (C) ±5
   (D) undefined

10. Solve: $\sqrt{(x + 7)} = x - 5$
   (A) $x = 9$
   (B) $x = 2$
   (C) $x = 9$ or 2
   (D) No solution

Section B: Fill in the Blanks (1 mark each)

11. If $x + y = 7$ and $xy = 12$, then $x^2 + y^2 =$ ___
12. If $x^2 - 6x + 8 = 0$, then $x =$ ___ or ___
13. $(a - b)^3 = a^3 - 3a^2b + 3ab^2 -$ ___
14. If $a + b + c = 0$, then $a^3 + b^3 + c^3 =$ ___
15. $(x + 1/x)^2 - (x - 1/x)^2 =$ ___
16. Rationalize: $1/(\sqrt{3} + 1) = (\sqrt{3} - 1)/$___
17. $\sqrt[3]{(-64)} =$ ___
18. HCF × LCM of 12 and 18 = ___

Section C: Short Answer Questions (2 marks each)

19. Prove: $(a + b)^3 - (a - b)^3 = 2b(3a^2 + b^2)$

20. Solve: log■(x - 1) + log■(x + 1) = 3

21. Factorize: $8x^3 - 27$

22. If α and β are roots of $x^2 - 5x + 6 = 0$, find $\alpha^3 + \beta^3$

23. Simplify: (x■ - 16)/(x^2 + 4)

24. Solve: $x^2 - 7x + 12 = 0$

25. If $x = 2 + \sqrt{3}$, find $x + 1/x$

--- End of Worksheet ---